

Multidimensional multiphysics modeling of TRISO particle fuel with SiC/ZrC coating using modified fission gas release model

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ABSTRACT

TRISO particle is adopted in the design of various types of reactors and accident tolerant fuel (ATF) due to its excellent accident tolerant performance under high temperature and other severe conditions. This study presents a multidimensional thermomechanical analysis of TRISO particle fuel with SiC/ZrC coating under typical operation conditions in reactor. Key fuel performance phenomena were considered including thermal performance (heat generation, conduction, irradiation gap heat transfer, etc.), mechanics (thermal expansion, elastic strain, densification, fission product swelling strain, contact, and thermal, irradiation creep, etc.), also irradiation behaviors (burnup, fission gas production and release, gap/plenum pressure with plenum volume, etc.). All these models have been integrated and solved via finite-element method. As for results, several complicated models were implemented and proved to successfully predict the thermomechanical behaviors of TRISO coated fuel compared with the existing experimental or numerical results. An enhanced fission gas behavior model considering grain boundary diffusion was established, which could complete detailed calculation on fission gas release at both grain size and fuel particle pellet scales. Based on the preliminary thermomechanical analysis, in terms of structural strength and heat transfer efficiency, ZrC appears to be a candidate material of SiC. Moreover, failure mechanisms and effective thermal conductivity have been analyzed using the current modified analysis technique.

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1. Introduction

As a newly developed fuel concept, TRistructural-ISotropic (TRISO) coated fuel particle (Powers and Wirth, 2010) has been accepted by several advanced fuel or types of reactors as a promising fuel type for effective radioactivity release barrier and low probability of failure, including fully ceramic microencapsulated (FCM) fuel (Brown et al., 2013), M3 fuel (Terrani et al., 2012), high temperature gas reactors (HTGRs) and thorium-fueled pebble bed fluoride saltcooled high-temperature reactor (PB-TFHR).

TRISO fuel particles are typically 800–1000 μm in diameter (Powers and Wirth, 2010). As shown in Fig. 1, a TRISO particle consists of a fuel kernel at the center surrounded successively by several layers, the composition and function (Powers and Wirth, 2010; Hales et al., 2013) of each layer are shown in Table 1. It's worth

mentioning that as the primary pressure boundary, SiC layer also can be replaced by zirconium carbide (ZrC), but performance capability of ZrC needs to be demonstrated. Few researches have been carried out on ZrC-coated particle (TRIZO) fuel (Portera et al., 2013; McEachern et al., 2012). Irradiation tests (Ogawa and Fukuda, 1985) indicated that ZrC may show better performance against kernel migration failure and fission products chemical corrosion than SiC. However, quantitative and accurate simulation study on TRIZO particle is far from sufficient, especially by multi-scale and multi-physics coupled modeling involving thermal performance and mechanical analysis is still a gap.

Numerous numerical models have been developed for the past few years to analyze TRISO fuel behavior under normal operation and accident conditions. The summary of several major TRISO coated fuel performance models and codes have been described in several literatures (Powers and Wirth, 2010; Demkowicz et al., 2019). Besides, other programs based on finite element method have also been applied to the modeling of different types of fuel

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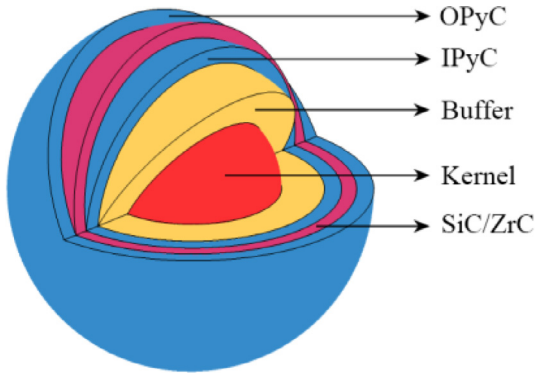


Fig. 1. Cutaway spherical geometry of a TRISO-coated fuel particle.

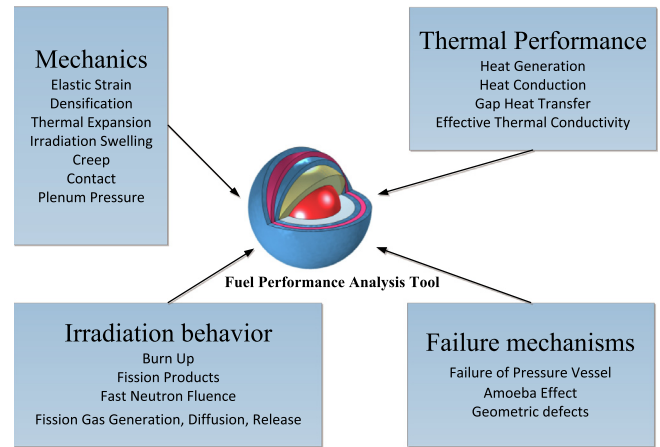


Fig. 2. Modeling structure of the TRISO fuel analysis tool.

Table 1
Composition and function of each layer in TRISO-coated fuel particle.

Layer	Material	Function
Kernel	UN/UO ₂ / UC ₂ /ThO ₂	Generate heat
Porous Buffer	Pyrolytic Carbon	- Provides a substrate for buffer layer - Storage fission gas and carbon oxide - Buffer the damage of fission fragments to the IPyC layer
Inner Pyrolytic Carbon	Pyrolytic Carbon	- Absorb the swelling of the fuel kernel - Keep chlorine and hydrochloride from permeating the fuel kernel - Prevent corrosion of SiC layer by fission products
SiC/ZrC	CVD-SiC/ ZrC	- Substrate for PyC layer - Primary containment for gaseous and solid fission products - Withstand internal pressure to maintain structural integrity
Outer Pyrolytic Carbon	Pyrolytic Carbon	- Protect SiC layer during manufacturing - Apply pressure to SiC layer

recently, Hales et al. (2013) analyzed the TRISO-coated particle fuel using BISON code based on MOOSE (Multiphysics Object-Oriented Simulation Environment) platform, Li and Shirvan (2019) studied the thermomechanical performance of a two-layer SiC cladding fuel using ABAQUS, Liu et al. (2020) constructed thermomechanics models to predict thorium-based fuel performance with a two-layer SiC cladding using COMSOL Multiphysics platform. However, fission gas release model used in the above work are found not quite comprehensive, which need to be improved. Also, the swelling as a result of gaseous fission products always adopts the empirical relations from theory manual like MATPRO (Allison et al., 1993), which can be replaced by mechanistic model for fuel swelling by considering intragranular and intergranular fission gas bubbles behavior.

In this paper, a multidimensional multiphysics fuel performance analysis technique based on the framework of COMSOL Multiphysics and its user-defined physical models has been developed to simulate the performance of TRISO Particle Fuel. COMSOL can be modified through the Application Programming Interface (API) or Graphical User Interface (GUI) scripting via JAVA or MATLAB, and by means of finite element analysis (FEA) solver in COMSOL, multiple physics have been taken into account in a fully coupled way. The modeling structure is shown in Fig. 2. An enhanced fission gas release model including fission gas diffusion in grain boundary, resolution, hydrostatic pressure, grain face bubbles coalescence and a gaseous swelling model by fission gas are established. Moreover, SiC and ZrC are both considered as the primary pressure boundary for the TRISO-coated fuel, and respective

probability of failure is compared in different failure models including kernel migration and geometric defects. What's more, effective thermal conductivity of different fuel types with different fuel and matrix was analyzed. This fuel performance analysis technique was validated with experimental data and other existing models or fuel performance analysis codes.

2. Material properties

2.1. TRISO fuel properties

2.1.1. Thermal conductivity

In analysis of fuel performance, the thermal conductivity of fuel kernel should be considered with respect to burnup and temperature. The fuel material considered is UO₂. Lucuta (1996) proposed the most acknowledged thermal conductivity model so far, on the basis of Harding and Martin (1989) relation.

$$k = k_0 \cdot FD \cdot FP \cdot FM \cdot FR \quad (1)$$

where k_0 denotes unirradiated thermal conductivity; FD denotes dissolved fission products correction; FP denotes precipitated fission products correction; FM denotes porosity correction; FR denotes radiation damage correction.

The thermal conductivity of porous pyrolytic carbon (buffer) and dense isotropic PyC layers (Petti et al., 2004) is 0.5 W/(m·K) and 4 W/(m·K), respectively.

The unirradiated conductivity of SiC as a function of the temperature (Snead et al., 2007) can be described with the following correlation ($T > 300$ K):

$$k_{non,irr} = (-3.0 \times 10^{-4} + 1.05 \times 10^{-5}T)^{-1} \quad (2)$$

Neutron irradiation introduces defects in the SiC material resulting in an increase of the thermal resistance. The conductivity of irradiated SiC can then be described (Petti et al., 2004) as:

$$k_{SiC} = (1/k_{non,irr} + 1/k_{rd})^{-1} \quad (3)$$

The thermal resistance ($1/k_{rd}$) due to radiation is the interpolation of 0.031 and 0.040 at fast neutron fluence levels of 1.0×10^{25} and 5.0×10^{25} n/m², respectively. The thermal conductivity of SiC as a function of temperature and burn up is shown in Fig. 3.

The thermal conductivity of ZrC is fitted from the measured data in Taylor's (1962) research. The polynomial regression formed equation, as shown in Eq. (4), was acquired by fitting function organizer in Origin. The fitting result is shown in Fig. 4.

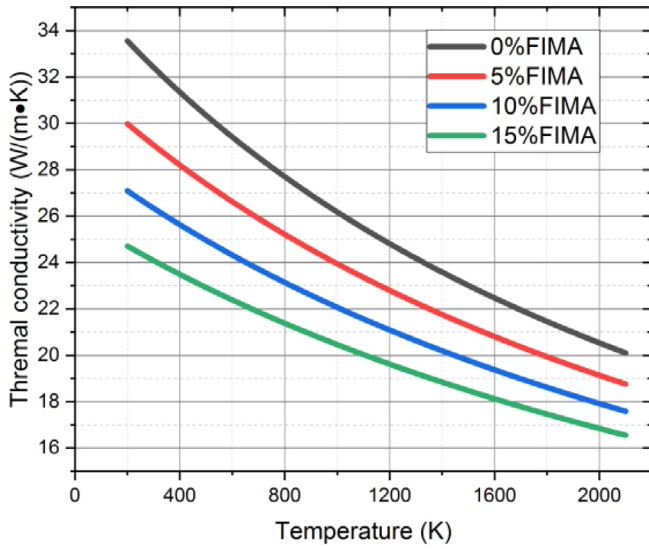


Fig. 3. Thermal conductivity of SiC.

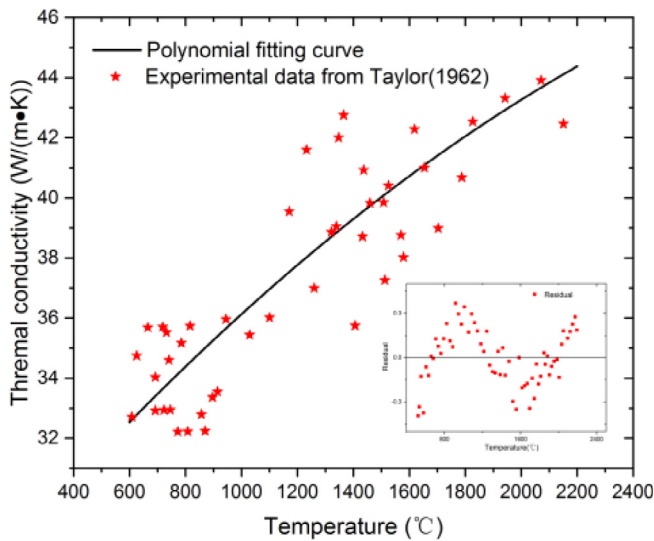


Fig. 4. Thermal conductivity of ZrC.

$$k_{ZrC} = 26.36 + 1.11 \times 10^{-2}T - 1.31 \times 10^{-6}T^2 \quad (4)$$

where T denotes temperature in the unit of $^{\circ}\text{C}$

2.1.2. Density and Specific heat

Density and Specific heat used for the particle materials are specified in Table 2 (Powers and Wirth, 2010). Density of all layers was assumed constant. The MATPRO-11 correlation (Carbajo et al.,

2001) for the heat capacity of Urania was used in the simulation. The temperature-dependent correlation for the heat capacity of SiC was defined from Xu et al. (2014) work, where T denotes temperature in the unit of K; Y is oxygen-to-metal ratio; R denotes universal gas constant; θ the Einstein temperature(K); E_D denotes activation energy for Frenkel defects (J/mol).

2.1.3. Young's modulus, Poisson's ratio and thermal expansion coefficient

For simplicity, thermal expansion coefficient (CTE) and key elastic properties including Young's modulus and Poisson's ratio for TRISO particle materials were assumed constant. Parameters used in the simulation were given in Table 3 (Powers and Wirth, 2010; Carbajo et al., 2001; Wongsawaeng, 2010).

2.2. Matrix properties

The TRISO particles are embedded in different matrix to form different fuel pellets. In a typical ATF for LWRs or SMRs, the matrix could be the nano infiltration transient eutectic SiC (NITE-SiC) in FCM fuel, or Zircaloy in metal matrix microencapsulated fuel.

In this work, the effective thermal conductivity of ceramic/metal matrix microencapsulated fuel is investigated, as an exploratory assessment of the impact of irradiation temperature, burnup, and matrix materials. The properties used in matrix materials are shown in Table 4 (Schappela et al., 2018; IAEA-TECDOC, 2006; Allison et al., 1993), where T denotes temperature in the unit of K.

3. Model implementation

3.1. Burnup

Burnup calculation in a TRISO fuel is induced based on the Olander (1976) equation as following:

$$\beta = \frac{\bar{F} t}{N_f} \quad (5)$$

where β denotes the burnup in %FIMA (fission of initial metal atoms), $\bar{F}$ denotes the volumetric fission rate in a unit of fission/ m^3/s , t denotes the time(s), and N_f the initial density of heavy metal atoms in the fuel (kg/m^3).

Fast neutron fluence is computed by time-integrated fast neutron flux:

$$\Phi = \int \Phi dt \quad (6)$$

where Φ is the fast neutron fluence (n m^{-2}), t the time(s) and Φ is the fast neutron flux ($\text{n m}^{-2} \text{s}^{-1}$).

3.2. Heat generation and conduction model

The heat transfer model was developed based on the existing module "Heat transfer in Solids" included in COMSOL. The basic heat conduction equation can be expressed as:

Table 2
Thermal properties and correlations used in TRISO-particle materials.

Material	Density (kg/m^3)	Specific heat ($\text{J}/\text{kg K}$)
UO ₂	11,000	$\frac{K_1 \theta^2 \exp(\theta/T)}{T^2 [\exp(\theta/T) - 1]^2} + K_2 T + \frac{Y K_3 E_D}{2RT^2} \exp(E_D/RT)$
Buffer	1000	720
PyC	1900	720
SiC	3180	$\begin{cases} 926.65 + 0.3772T - 7.9259 \times 10^{-5}T^2 - 3.1946 \times 10^{-7}T^{-2} & 200\text{K} < T < 2400\text{K} \\ 1365.54 + 1.379 \times 10^{-3}T & 2400\text{K} < T < 2800\text{K} \end{cases}$
ZrC	6590	485

Table 3
Elastic properties and thermal expansion coefficient (CTE) for the particle materials.

Material	Young's modulus (GPa)	Poisson's ratio	Thermal expansion coefficient (/K)
UO ₂	220	0.34	10.0×10^{-6}
Buffer	20	0.23	5.7×10^{-6}
PyC	47	0.23	5.7×10^{-6}
SiC	340	0.13	4.9×10^{-6}
ZrC	400	0.25	7.1×10^{-6}

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q \quad (7)$$

where ρ , C_p and k denote density, specific heat capacity and thermal conductivity, respectively. Q accounts for heat source.

The gas conductance h_{gas} is described using the form proposed by Ross and Stoute (1962):

$$h_{\text{gas}} = \frac{k_{\text{gas}}}{d_{\text{gas}} + C_r(r_b + r_i) + g_b + g_i} \quad (8)$$

where k_{gas} is the conductivity of the gas in the gap, computed via MATPRO (Allison et al., 1993); d_g is the gap width (computed dynamically via irradiation deformation), C_r is a roughness coefficient with r_b and r_i the roughnesses of the two surfaces, and g_b , g_i are jump distances at the two surfaces.

The radiant conductance is described as:

$$h_r = \frac{\sigma(T_b^2 + T_i^2)(T_b + T_i)}{1 + 1/\varepsilon_b + 1/\varepsilon_i - 1} \quad (9)$$

where σ is the Stephan-Boltzman constant, ε_b and ε_i are the emissivities of the radiating surfaces, T_b and T_i are the temperatures of the radiating surfaces.

3.3. Deformation mechanics

3.3.1. Elastic deformation

The deformation behavior model was developed based on the ‘‘Structural Mechanics’’ module included in COMSOL. The Cauchy’s equation is described by:

$$-\nabla \cdot \sigma = F_v \quad (10)$$

where σ is the Cauchy stress tensor and F_v is the body force per unit volume. The stress is calculated with a liner elastic constitutive model $\sigma = [C][\varepsilon_e]$ (C is the material matrix, ε_e is the elastic strain vector). In a geometrically linear analysis, the elastic strain is computed by a straightforward subtraction of the inelastic strain:

$$\varepsilon_e = \varepsilon - \varepsilon_{\text{inel}} \quad \varepsilon = \frac{1}{2} [\nabla u + (\nabla u)^T] \quad \varepsilon_{\text{inel}} = \varepsilon_0 + \varepsilon_{\text{ext}} + \varepsilon_{\text{th}} + \varepsilon_{\text{cr}} \quad (11)$$

where ε denotes the total strain, $\varepsilon_{\text{inel}}$ denotes the inelastic strain, u denotes the displacement vector, ε_0 is initial strain, ε_{ext} is external strain, ε_{th} denotes thermal strain and ε_{cr} accounts for creep strain. The irradiation deformation was added in constitutive equations via weak form in the user-defined module. Also, in mechanics mod-

ule, we used the COMSOL built-in contact module including penalty module and sugmented lagrangian methods.

3.3.2. Fuel Densification, solid fission product swelling and creep

Fuel densification is computed using the ESCORE empirical model (Rashid et al., 2004), which is:

$$\varepsilon_D = \Delta \rho_0 \left(e^{\left(\frac{Bu_{\text{in}}(0.01)}{C_D Bu_D} \right)} - 1 \right) \quad (12)$$

where ε_D denotes the densification strain, $\Delta \rho_0$ is the total densification which can occur, Bu_D is the burnup when densification is completed. For temperatures below 750 °C the parameter C_D is given by 7.2–0.0086 ($T-25$); above 750 °C is 1.0 (T in °C).

Volumetric strain as a result of solid fission product swelling is expressed using empirical relations from MATPRO (Allison et al., 1993):

$$\frac{\delta \varepsilon_{\text{sw-s}}}{\delta t} = 5.577 \times 10^{-5} \rho \frac{\delta Bu}{\delta t} \quad (13)$$

where $\varepsilon_{\text{sw-s}}$ is the volumetric solid swelling, Bu is the burnup in fissions/atoms-U, and ρ the density (kg/m^3).

While swelling due to gaseous fission products is contributed by both the intragranular and intergranular fission gas bubbles. Thus the gaseous swelling was coupled with the fission gas behavior in section 3.4.3.

The creep model for UO₂ has been substituted using The MATPRO FCREEP model (Allison et al., 1993), given as:

$$\dot{\varepsilon} = \frac{A_1 + A_2 \dot{F}}{(A_3 + D)G^2} \sigma e^{\left(\frac{-Q_1}{RT} \right)} + \frac{A_4}{(A_6 + D)} \sigma^{4.5} e^{\left(\frac{-Q_2}{RT} \right)} + A_7 \dot{F} \sigma e^{\left(\frac{-Q_3}{RT} \right)} \quad (14)$$

where $\dot{\varepsilon}$ is the creep rate (1/s), σ is the effective stress (Pa), T is the temperature (K), D is the fuel density (in percent of theoretical), G is the grain size (μm), $\dot{F}$ is the volumetric fission rate (fissions/ m^3/s), Q_i are the activation energies (J/mol), R is the universal gas constant and A_1 to A_7 are material constants.

3.3.3. Irradiation deformation of coated layers

The irradiation-induced deformation of Buffer, PyC and CVD-SiC are modeled in this work. The irradiation strain for the buffer as a function of fast neutron fluence layer (Hales et al., 2013) is described as:

$$\dot{\varepsilon}_r = \dot{\varepsilon}_\theta = -0.176 e^{(-1.75\Phi)} \quad (15)$$

where $\dot{\varepsilon}$ is strain in units of 1/s and Φ the fast neutron fluence in units of 10^{25} n/m^2 . The subscript of ‘ r, θ ’ denotes the radial and tangential directions.

Eqs. (24) and (25) presents the correlations used to simulate the irradiation-induced strain of dense pyrolytic carbon (Hales et al., 2013):

$$\dot{\varepsilon}_r = -0.077 e^{(-\Phi)} + 0.031 \quad (16)$$

$$\dot{\varepsilon}_\theta = -0.036 e^{(-2.1\Phi)} - 0.01 \quad (17)$$

Table 4
Properties used in matrix materials.

Material Properties	NITE-SiC	Zr
Thermal conductivity (J/kg K)	Eq. (8)	$11.498 + 4.6765 \times 10^{-3} T + 2.761 \times 10^{-6} T^2 + 2.2147 \times 10^{-6} T^3$
Density (kg/m^3)	3210	6595
Specific heat (J/kg K)	1200	256
Young's modulus (GPa)	400	108
Poisson's ratio	0.2	0.32
Thermal expansion coefficient (/K)	4.5×10^{-6}	7.1×10^{-6}

The irradiation creep of pyrolytic carbon materials including Buffer and PyC (Schappela et al., 2018) are given as:

$$\dot{\epsilon}_{cr} = K[\sigma_1 - \nu_c(\sigma_2 + \sigma_3)]\dot{\Phi} \quad (18)$$

$$K = 2K_0[1 + 2.38(1.9 - \rho_0)] \quad (19)$$

$$K_0 = 1.996 \times 10^{-29} - 4.415 \times 10^{-32}T + 3.6544 \times 10^{-35}T^2 \quad (20)$$

where K is the creep constant, ρ_0 is the initial density in g/cm^3 , σ_i as one component of the principal stress, ν_c is the Poisson ratio for creep, T is the temperature in $^\circ\text{C}$, and $\dot{\Phi}$ is the fast neutron flux ($\text{n m}^{-2} \text{s}^{-1}$).

Eq. (29) provides the modeled irradiation creep behavior (Schappela et al., 2018) for SiC materials:

$$\dot{\epsilon}_{cr} = K\dot{\Phi}\sigma_e \quad (21)$$

where $\dot{\epsilon}_{cr}$ is creep coefficient (s^{-1}), $\dot{\Phi}$ is the neutron flux, and σ_e the effective stress.

3.4. Fission gas behavior

The fission gas behavior model is added on the basis of Cui's (2012, 2013) work. Different from the Forsberg and Massih (1985) two-stage model, the more detailed grain boundary diffusion model is considered in this work. The following phenomena are taken into account in our model, including the gas atom diffusion, resolution, grain boundary diffusion, bubble growth, grain face bubbles coalescence, saturation and the effects of the grain sizes and the external pressure.

In general, fission gas atoms generated in grain interior diffuse to grain boundary, and from grain boundary to intergranular bubble, and then from intergranular bubble to the free volume in succession. Under this assumption, the fission gas release (FGR) is calculated when grain surface gets saturated and any additional fission gas at the grain surface will be released to the free volume.

3.4.1. Intragranular model

The intragranular fission gas behavior model is calculated by Cui's (2012) work. The governing equation with a continuously distributed source in spherical coordinates for fission gas diffusion in the irradiation of UO_2 is:

$$\frac{\partial \psi}{\partial t} = D_{eff} \cdot \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + Y \cdot \dot{f} \quad (22)$$

where ψ is total concentration of both dissolved gas and the gas trapped in intragranular bubbles contained in each grain (m^{-3}), t the time(s), D_{eff} is the effective diffusion coefficient in units of $\text{m}^2 \text{s}^{-1}$, r is the radial coordinate in a spherical grain(m), Y is the number of the gas atoms created per fission, $\dot{f}$ is the fission rate ($\text{m}^{-3} \text{s}^{-1}$).

The gas concentration in the grain boundary N_{min} is given as:

$$N_{min} = \min(N(t), N_s) \quad (23)$$

where $N(t)$ represents the total gas atom concentration generated inside the grain to its boundary, which can be determined by the total mass balance of the generated gas atoms:

$$\frac{4}{3}\pi a^3 \int_0^t Y \dot{f} dt = \int_0^a \psi 4\pi r^2 dr + 4\pi a^2 \frac{N(t)}{2} \quad (24)$$

N_s is the saturation concentration. If $N(t) > N_s$, fission gas release will happen. Parameters used are taken as the same as in Denis's (1996) and Cui's (2012) work.

3.4.2. Grain boundary fission gas and intergranular bubbles

The intergranular model established by Kogai (1997) features the treatment of the grain boundary assumed to consist of two zones for solute gas atoms and intergranular bubbles. Conservation of the gas in grain boundary and intergranular bubble is written as:

$$\frac{dC_{gb}}{dt} = (1 - F_c)J_{intra} - J_{gb-bl} \quad (25)$$

$$\frac{dC_{bl}}{dt} = F_c J_{intra} + J_{gb-bl} - J_{release} \quad (26)$$

where, C_{gb} is the gas amount in grain boundary (m^{-3}), C_{bl} is the gas amount in intergranular bubble (m^{-3}), J_{intra} is the gas flow from grain interior to the grain boundary ($\text{m}^{-3} \text{s}^{-1}$), J_{gb-bl} is the gas flow from grain boundary to intergranular bubble ($\text{m}^{-3} \text{s}^{-1}$), $J_{release}$ is the gas flow from intergranular bubble to the free volume ($\text{m}^{-3} \text{s}^{-1}$), and F_c is the bubble coverage of grain boundary.

The bubble radius can be calculated by Matthews's (1980) work:

$$\frac{dr_{bl}}{dt} = \frac{\pi \delta_b D_{gb}'' \Omega}{4r_{bl}^2 kT} (P_{bl} - P_h - 2\gamma \sin \theta / r_{bl}) k_f^2 \quad (27)$$

where δ_b is the grain boundary width, D_{gb}'' is the vacancy migration coefficient in the boundary, P_h is hydrostatic pressure, Ω is atomic volume, k_f^2 represents sink strengths of face bubbles.

Veshchunov and Tarasov (2009) proposed an improved bubbles coalescence model in 2008:

$$\frac{dF_c}{dt} = \frac{n_{bl}}{1 + 2F_c} \frac{d\pi r_{bl}^2}{dt} \quad (28)$$

All these parameters can be found in Kogai (1997), Denis's (1996), Cui (2012, 2013), Matthews and Wood (1980), Veshchunov and Tarasov (2009).

3.4.3. FGR and gaseous fission products swelling

Based on the above mathematical expressions of the model, the fractional fission gas release (Cui et al., 2012) is defined as:

$$FGR = \frac{\max \left\{ \int_0^t Y \dot{f} dt - a^{-3} \int_0^a 3\psi r^2 dr - C_{gb} - C_{bl}, 0 \right\}}{\int_0^t Y \dot{f} dt} \quad (29)$$

The intergranular swelling model is adopted according to the study of Forsberg and Massih, which is based on the assumptions: (1) The number of the intergranular bubbles is constant. (2) Their radii increase until the grain boundary gets saturated. The fractional swelling due to fission gas is thus given by:

$$\frac{\Delta V}{V} = \frac{3}{2a} \left(\frac{r_b}{r_{bs}} \right)^3 N_s \left(B + \frac{k_b T}{P_{ext} + 2\gamma / r_{bs}} \right) \quad (30)$$

Eqs. (22)–(30) of fission gas behavior are induced in COMSOL via weak form partial differential equation (PDE) and boundary ordinary differential equation (ODE) through a one dimensional model. Here, a set of these equations is discretized for time by the backward Euler method. The amount of gas released and swelling per unit volume can be calculated at grain size level, and then by integrating over the whole UO_2 pellet, the fission gas release and swelling at pellet scale could be obtained. Fig. 5 shows the schematic diagram of multi-scale coupling model for fission gas behavior.

3.5. Plenum pressure

It is assumed that all oxygen released is combined with carbon in the buffer layer to form CO. Carbon monoxide production in UO_2

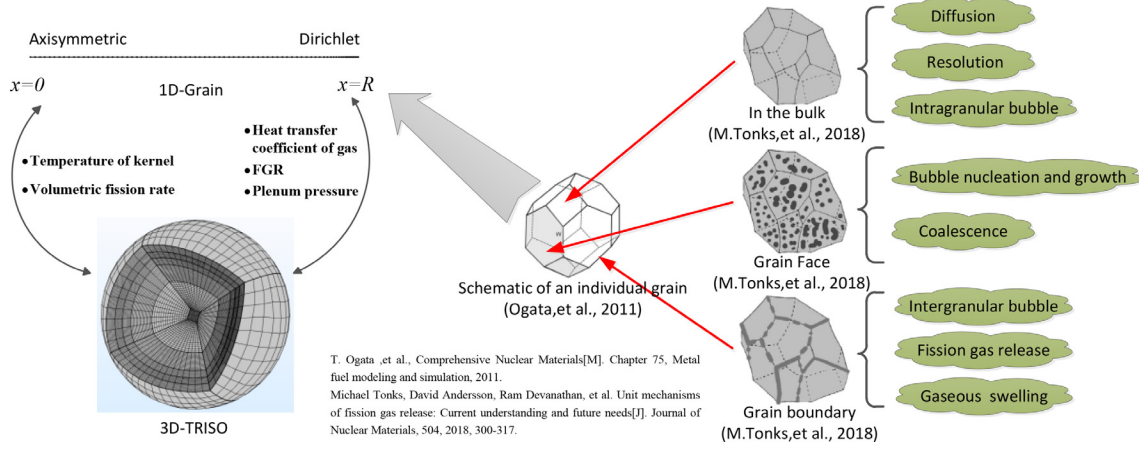


Fig. 5. Schematic diagram of multi-scale coupling model for fission gas behavior.

is computed using the empirical relation given by Proksch et al. (1982):

$$\frac{O}{F} = \frac{t^2}{(1.211 \times 10^{10}) 10^{\frac{8500}{T}}} \quad (31)$$

where O/F is the oxygen released (atoms/fission) with a maximum value of 0.4, t is the irradiation time (s) and T is the time-averaged particle surface temperature (K).

The pressure in the gap and plenum is computed based on the ideal gas law:

$$P = \frac{n_g R}{\frac{V_b}{T_b} + \frac{V_g}{T_g}} \quad (32)$$

where P is pressure, n_g is the moles of gas, R is the ideal gas constant, T_b is the temperature of buffer, T_g is the temperature of buffer and inner IPyC, and V_b is gas-filled volume in buffer (related to porosity) and V_g is the volume of gap (related to burnup).

3.6. Failure mechanisms

TRISO fuel is the first barrier to the release of fission products, to some extent, the performance of the coated fuel particles determines the safety of the reactor. It is necessary to study the failure mechanisms using numerical method since traditional irradiation experiments have long period and high cost. In this study, the main mechanisms causing the failure of coated fuel particles including pressure vessel failure, amoeba effect and geometric defects were modeled and analyzed.

3.6.1. Failure of gas pressure vessel

Fission gases and CO are released from kernel under irradiation, the inner pressure build-up increases with burnup and temperature and leads to tensile stresses on the dense coating layers of the particle. As the primary containment for fission products, the SiC/ZrC layer may fail when it cannot withstand internal pressure of coated fuel. The cumulative failure probability for coating layer is expressed by a Weibull distribution with Eq. (33)

$$\psi = 1 - \exp\{-\ln(2)(\sigma_t/\sigma_0)^m\} \quad (33)$$

where ψ is the failure probability, σ_t is the stress on the layer at irradiation time t , σ_0 is the fracture strength, m is the Weibull modulus.

3.6.2. Kernel migration (amoeba effect)

Kernel migration, namely amoeba effect, is simply defined as the movement of the kernel towards TRISO coated layers. The

driving force for the kernel migration is reaction between free oxygen in oxide based fuel kernels and carbon in the coating layers resulting in CO or CO₂ production. Results of irradiation tests gathered by Sawa and Ueta (2004) on kernel migration distances lead to the following correlation:

$$KMR = 2 \times 10^{-6} \exp\left(-\frac{14800}{T}\right) \frac{1}{T^2} \frac{dT}{dr} \quad (34)$$

where r is the radial distance in m, T is the temperature in Kelvins and KMR is the kernel migration rate in $\mu\text{m/s}$.

3.6.3. Geometric defects

Geometric defects could be a common phenomenon when manufacturing the fuel, which usually leads to aspherical geometry of TRISO particle. The purpose of this simulation is to explore whether stress concentration occurs in geometric defects area.

4. Code verification and demonstration problems

4.1. Code verification

The code was verified by IAEA HTGR benchmarks (IAEA, 2012), known as IAEA Coordinated Research Program (CRP-6), of which Case-8 was selected to verify fuel performance due to its realistic irradiation condition for TRISO-coated particles. Detailed boundary and initial conditions are given in IAEA (2012). Fig. 6 illustrates the tangential stress at the inner wall of the IPyC and SiC layers, compared to solutions by other codes extracted from the CRP-6 exercise, namely BISON, PARFUME, TRFUEL and STRESS3. As Fig. 6 shows, the calculations resulted by COMSOL are in good agreement with those of other programs. With the increase of fast fluence, IPyC layer is subjected to an increasing and then decreasing stress in the tangential direction, the inner surface of SiC layer is subjected to compressive stress, which gradually decreases due to internal pressure, and has been changed to tensile stress in end of the period. The sawtooth variation of stress corresponds to the periodic variation of temperature and internal pressure.

4.2. Demonstration problems

4.2.1. Radiation behavior analysis

The 1/8 of the TRISO fuel (shown in Fig. 7) in a three dimensional axisymmetric model was selected to analyze its thermo-mechanical performance. With a constant power 0.05 W and a fixed particle surface temperature (1500 K) over a irradiation

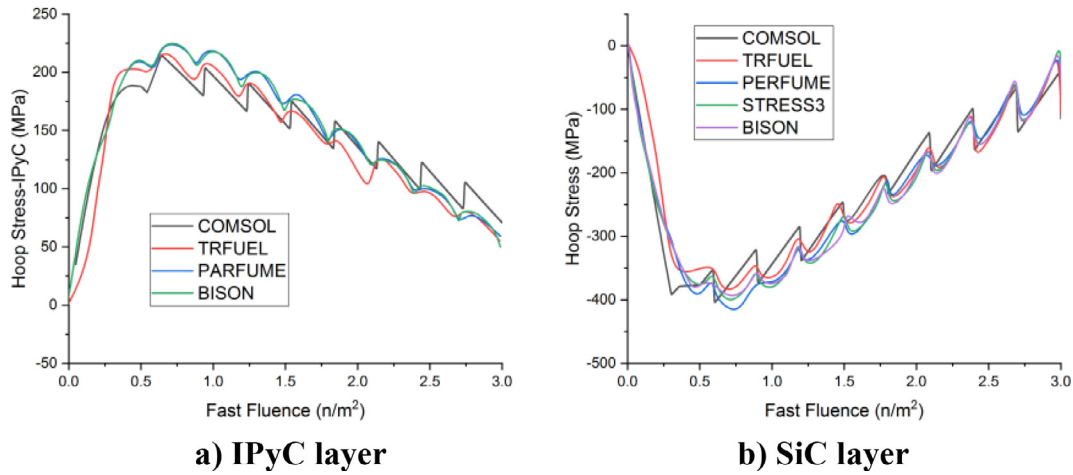


Fig. 6. Tangential stress of inner wall of IPyC and SiC layers.

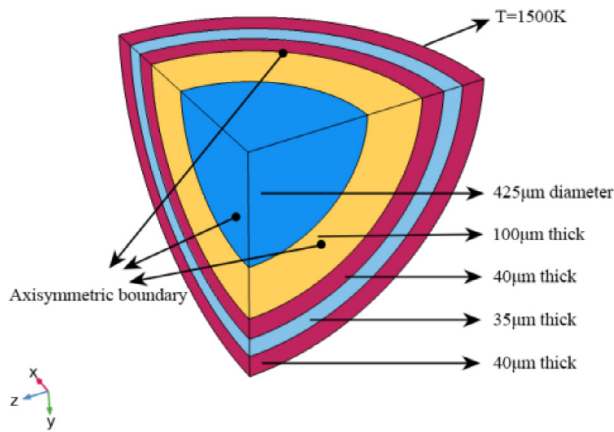


Fig. 7. Particle model assumed in COMSOL.

period of 76 Ms, the results were validated with Hales et al. (2013) research.

Fig. 8 performed an axisymmetric two dimensional model to simulate kernel migration. Migration direction is along the Z axis and a temperature gradient of 15000 °C/m is assumed, according to experiments carried out by Sawa and Ueta (2004). The kernel

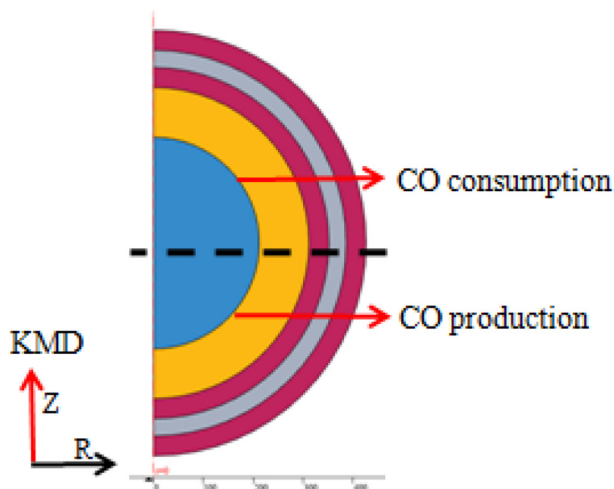


Fig. 8. The structure of fuel particle with kernel migration.

is assumed to preserve its spherical shape during the calculation, CO was assumed to be produced at bottom and consumed on top to represent the movement of the migrated kernel, and the fission gas model was coupled through boundary heat flux, with burnup up to 12%FIMA. The power is set to 0.05 W, outer surface temperature is 1500 K, The results were compared with (2008) (Gulol, 2008) qualitatively.

Geometric defects during manufacturing were simulated using an aspherical geometry in a 2D axisymmetry model as shown in Fig. 9. Half of the coated layers were assumed to be flat, the half-length of the flat section is 180–200µm. The same conditions described in Fig. 6 (temperature, power and irradiation) are considered here.

4.2.2. Effective thermal conductivity

A model for effective thermal conductivity of TRISO coated fuel pellets with different materials was developed. Simulation for three cases with different phase volumes (proportion of TRISO particles to the fuel volume) of 30%, 40%, 50% are performed, the representative fuel unit size of TRISO coated fuel pellets is shown in Table 5.

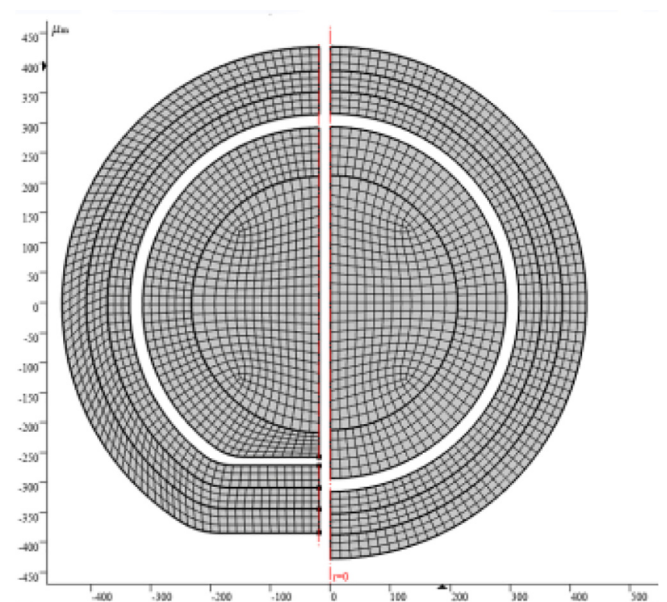


Fig. 9. Modeling of geometric defects.

Table 5
Geometric Dimensions of TRISO coated fuel pellets.

Phase volume of TRISO particle/%	Size of fuel unit/mm
30	514.62
40	467.57
50	434.05

The effects of burnup, fission gas released, appearance of buffer-IPyC gap, different matrix materials and irradiation temperature are preliminarily investigated. The temperature selected in the simulation ranges from 600 K to 2000 K, with burnup up to 12% FIMA. The simple cubic (SC) arrangement was adopted in this study, and the geometric model and boundary conditions of fuel volume is shown in Fig. 10.

5. Results and discussion

5.1. FGR and CO generation

1) At grain scale

In order to obtain the data of fission gas behavior and validate the model, Eqs. (22)–(30) in the previous section are solved simultaneously. The inputs to this model are fission rate, burnup, fuel temperature (acquired from result in 3-D TRISO) and hydrostatic pressure in pellet. It is noted that there is no obvious grain growth in the experiment, so the grain radius(a) in this work is taken as a constant of $10\mu\text{m}$ according to Denis, 1996, and the hydrostatic pressure $P_{\text{ext}} = 0$, other key parameters used in this model can be found in Cui et al., 2012; Matthews and Wood, 1980; and Veshchunov and Tarasov, 2009.

Fig. 11 shows the calculated FGR for different temperatures respectively. Results were compared with the experimental results measured by Olander (1976) extracted from Denis's (1996) study, also the numerical results calculated by Cui et al. (2012). Although the results agree not perfectly, it is a reasonable agreement considering the grain boundary diffusion model induced in this study.

Figs. 12 and 13 show the effects of the grain size and the external pressure on the FGR. It can be found in Fig. 12 that large grain size dramatically lowers the FGR. Fig. 13 shows that the FGR decreases to a certain extent by increasing the external pressure. Moreover, the saturation time gets longer as the hydrostatic pressure increases. There are no experimental data of grain radius or hydrostatic pressure available for comparison with the calculation by this model, however, the results show high consistency compared with Cui's numerical simulation.

2) At fuel scale

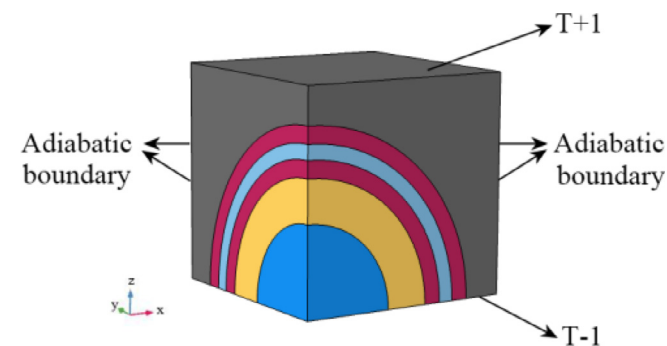


Fig. 10. Geometric model and boundary conditions of fuel volume.

By integrating the gas released in unit volume, The FGR at fuel pellet scale can be acquired. Fig. 14 shows the gas amount of fission gas and CO versus time under normal operation conditions. It is shown that the FGR calculated is in accordance with the numerical results calculated by Bison (Hales et al., 2013). The difference in fission gas released comes from the selection of grain boundary models.

Computed total gas pressure during the irradiation periods is shown in Fig. 15. The accumulation of gas in the Buffer-IPyC gap and Buffer layer causes the rise of internal pressure. The plenum pressure can reach 44 MPa when burnup is 12%FIMA. Because of more gas release calculated by COMSOL, the plenum pressure was about 8 MPa higher than result predicted by BISON.

5.2. Thermomechanical performance

Fig. 16 shows the radii change of the outer Buffer and inner IPyC due to the irradiation deformation, fission product swelling strain, densification and irradiation creep. At the late stage of burnup, due to radiation creep and radial expansion, the radius of IPyC layer decreased somewhat, while the radius of Buffer layer increased slightly due to radiation swelling of UO_2 , and the Buffer-IPyC gap width increased to about 21 μm , where fission gas and carbon monoxide can be stored. The two radius curves shows good agreement with results in Hales et al., 2013. It's worth mentioning that the radius difference of Buffer layer (due to gaseous swelling) is consistent with the FGR results described in section 5.1. Fission gas model adopted in this study calculated more gas release, also lead to the relatively less gaseous fission product swelling simultaneously.

Fig. 17 shows the computed temperature distribution of fuel particle with radius at burnups of 0% and 12% FIMA. At 0% FIMA the temperature profile is dominated by Buffer layer, since its larger thermal resistance. At later period, a gap between the Buffer layer and the IPyC layer was formed due to irradiation deformation, a steep gradient in the temperature was induced because of the significantly lower thermal conductivity of gases gathered in the gap.

Kernel migration (amoeba effect) is the movement of the kernel towards TRISO coated layers driven by the combination with free oxygen in UO_2 and carbon in the coating layers, here all oxygen released combined with carbon is assumed to form CO. With conditions defined in 4.2.1, the asymmetric temperature distribution along the Z axial direction is observed as a result of the kernel migration in Figs. 18 and 19. As shown in Fig. 18, the relatively low heat conduction of the CO results in the change of heat transfer on both sides of the kernel. For a single particle, the maximum kernel temperature is about 10–15 K greater than the unmigrated particle. Fig. 19 presents the heat flux proportion with increasing burnup. An increase in the proportion for the direction of kernel migration shows that the amoeba effect has a significant influence on the temperature distribution in the fuel kernel. However, according to Eq. (34), migration distance could be calculated as a function of time (burnup) based on a constant migration rate, the maximum migration distance is $55\mu\text{m}$ when outer surface is 1500 K, which clearly isn't enough to break through the porous Buffer or IPyC and destroy the structural integrity of the particle. On the other hand, for a 10–15 K rise in temperature, no noticeable impact on stress distribution of SiC or ZrC is observed. Therefore the effect of kernel migration itself is not so critical on fuel failure.

Fig. 20 shows the variation of maximum tangential (hoop) stress with time in main load bearing layers, which is SiC or ZrC. As fission gas is released and CO is produced, the compressive stress on the SiC and ZrC layer due to the shrinkage of the PyC layers gradually becomes tensile stress. Note that the tensile stress

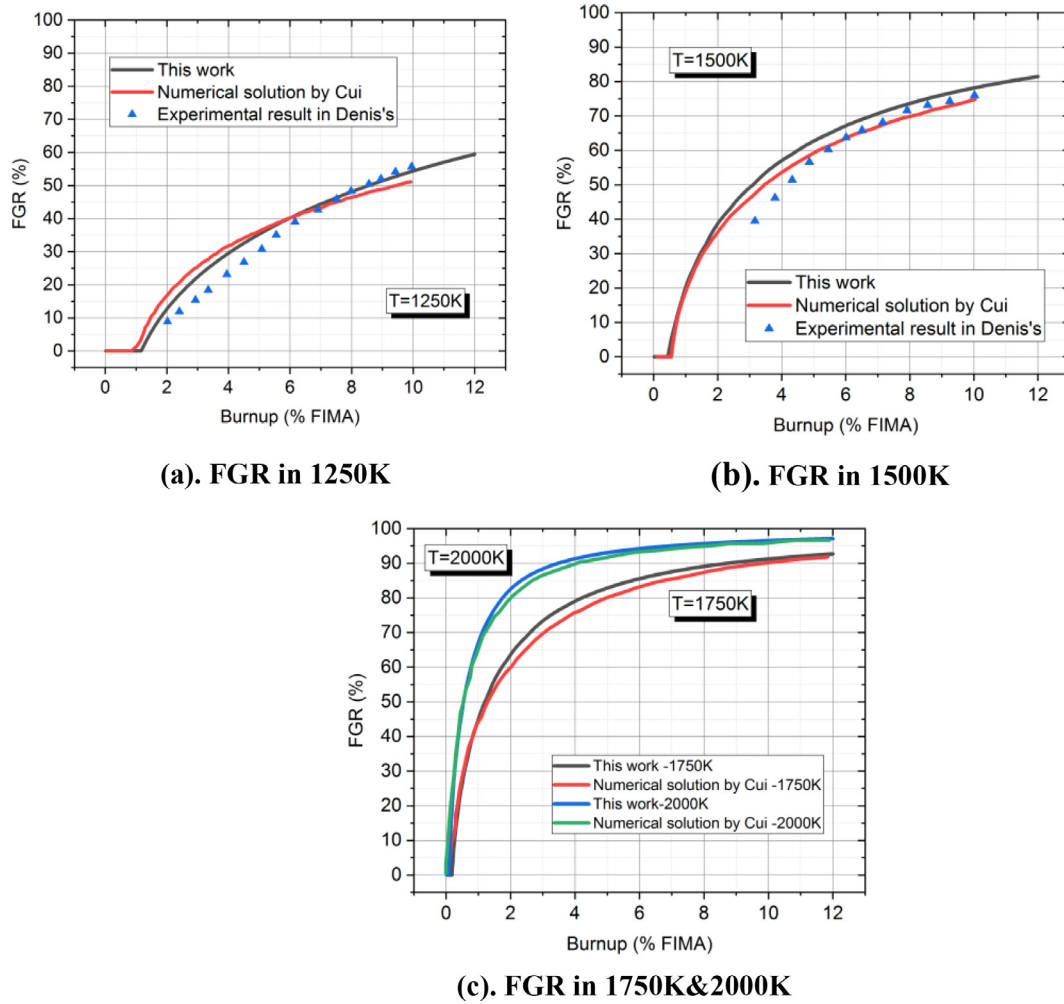
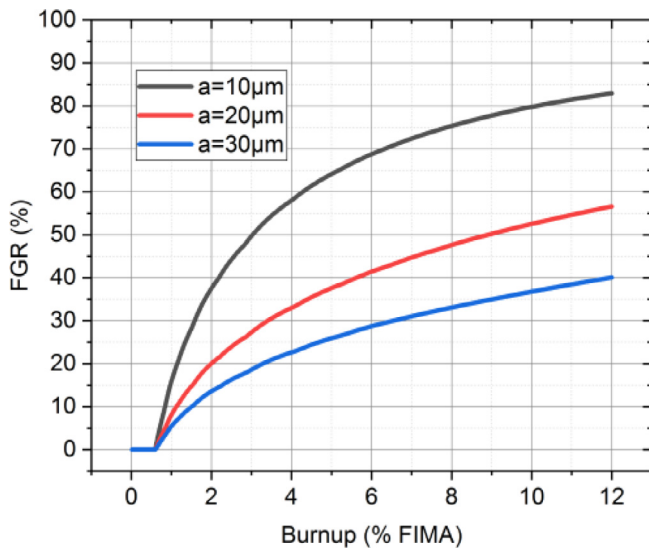
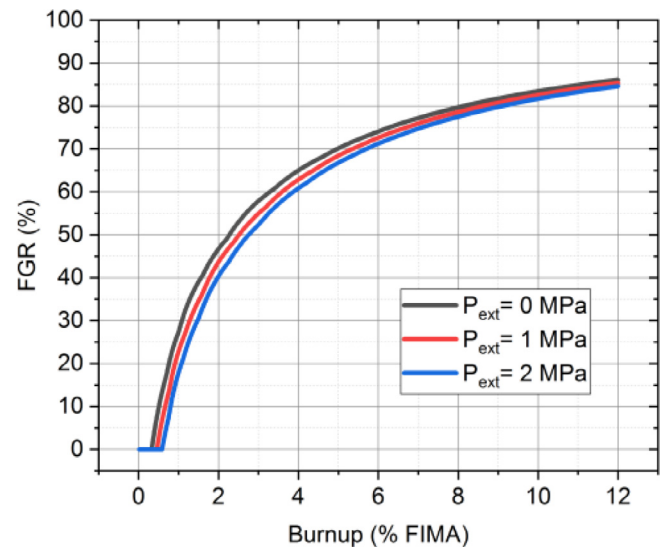


Fig. 11. Comparison of FGR with different temperatures.

Fig. 12. FGR with different grain radius ($P_{ext} = 0$, $T = 1500$ K).Fig. 13. FGR with different external pressure ($T = 1500$ K, $a = 10^{-5}$ m).

in the SiC layer is larger than ZrC, and the tension stress elevated significantly when the model has geometric defects.

When the tensile stress is greater than the fracture strength, failure of pressure vessel may occur in the SiC/ZrC layer, causing

the leakage of fission products. The failure probability of the SiC/ZrC is shown in Fig. 21. The ultimate bend strength for the SiC and ZrC are 480 MPa and 150 MPa, respectively. Weibull moduli for the SiC and ZrC are 8 and 6, respectively. The results imply that ZrC is

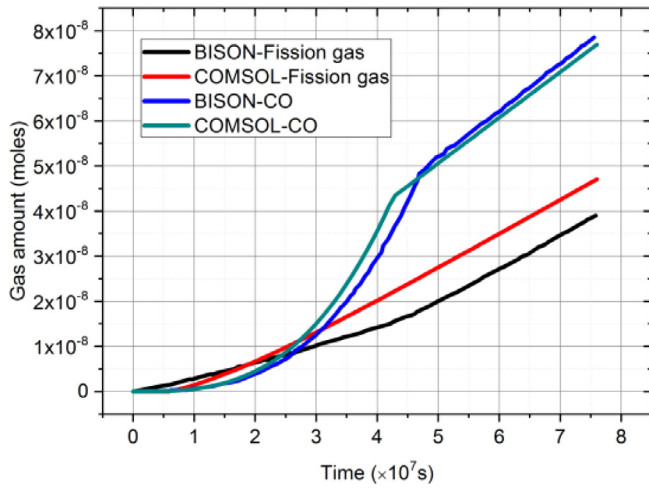


Fig. 14. FGR and CO generation in fuel particle.

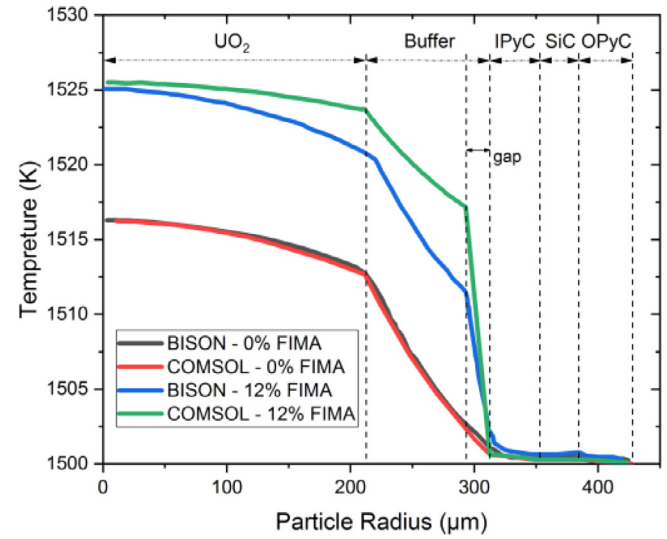


Fig. 17. Temperature distribution of fuel particle with radius.

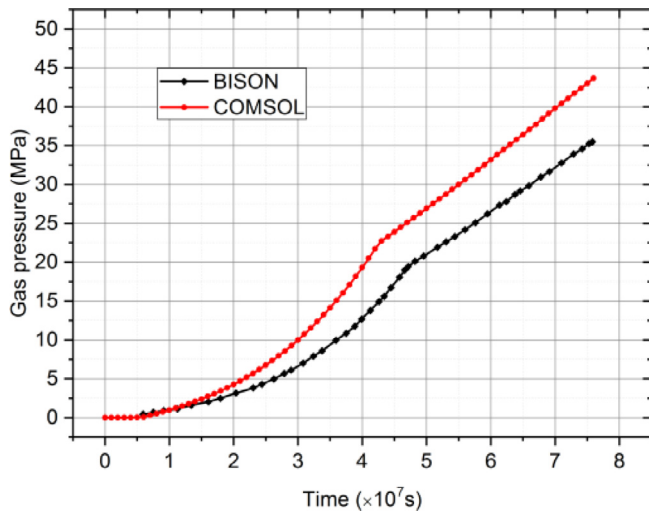


Fig. 15. Plenum pressure.

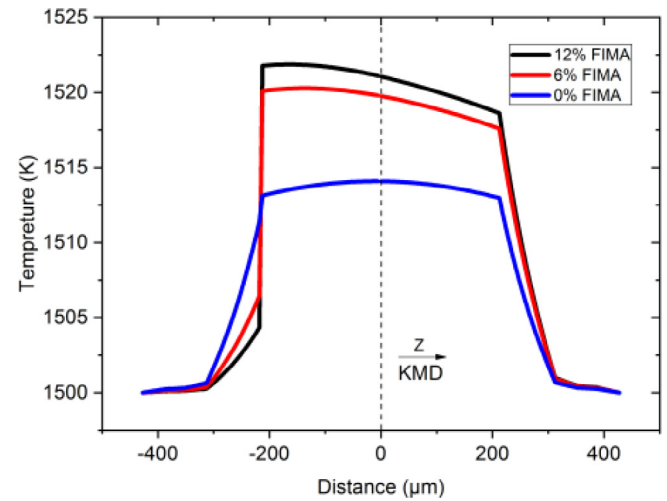


Fig. 18. Temperature distribution along the z-direction.

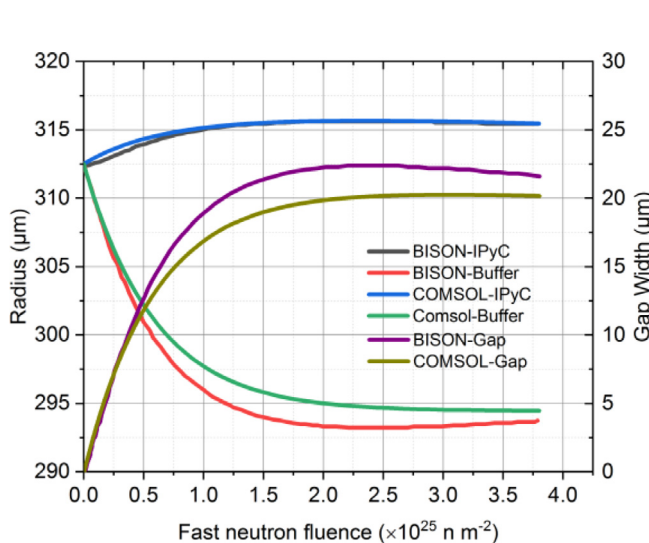
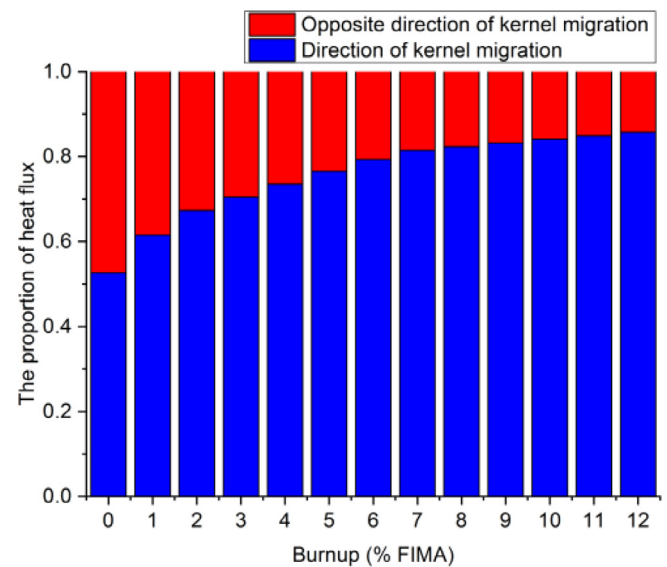
Fig. 16. The radii change of the outer Buffer and inner IPyC as a function of fast neutron fluence ($E > 0.1$ MeV).

Fig. 19. Proportion of heat flux versus burnup.

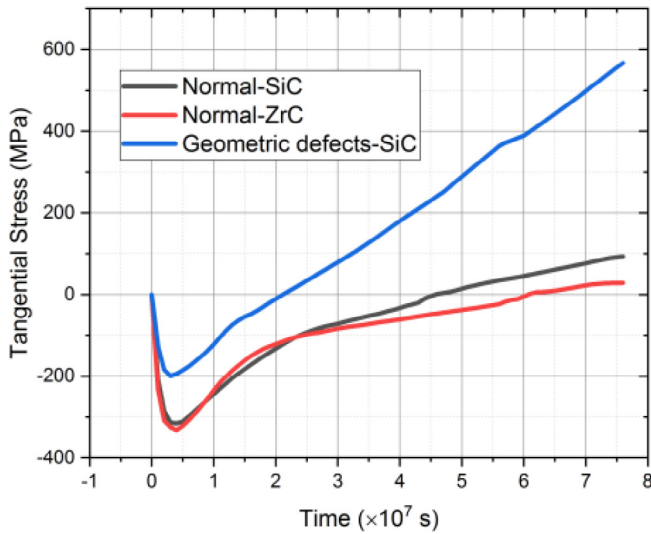


Fig. 20. Maximum tangential stress of inner SiC/ZrC.

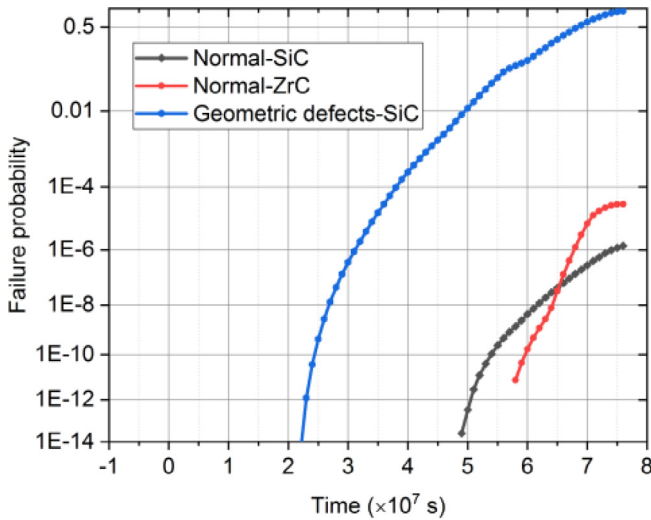


Fig. 21. Failure probability of the SiC/ZrC.

not as strong as SiC, however, if it's considered dangerous when the failure probability reach 10^{-4} (safety limit in some HTGRs), both of normal SiC or ZrC can be regarded as reliable under this situation.

Much higher failure probability shown in Fig. 21 indicates that geometric defects such as particle asphericity can be very crucial to the safety of the reactor core.

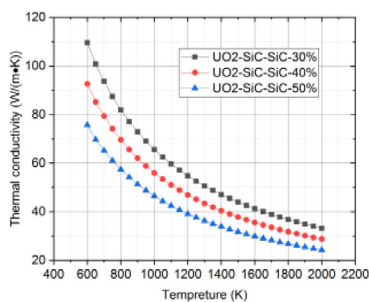
5.3. Effective thermal conductivity

The effective thermal conductivity under different kernel, pressure vessel and matrix materials are also studied whose results are shown in Figs. 22 and 23. Fig. 22 shows the effective thermal conductivity of unirradiated fuel vary with temperature. It can be seen that the effective heat transfer coefficient decreases with the increase of TRISO phase volume, and the change trend of effective thermal conductivity is more sensitive to the material of matrix (see Fig. 22(a) and (b)). The difference in thermal conductivity between SiC and ZrC as pressure vessel materials is not obvious, only that the UO_2 -ZrC-NITESiC fuel seems to be less sensitive to the change of TRISO phase volume (see in Fig. 22(c)).

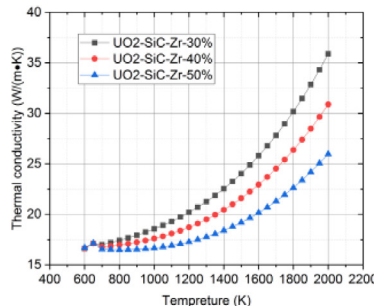
Furthermore, the irradiation deformation, appearance of Buffer-IPyC gap and fission gas behavior were taken into account in this model to investigate the effect of burnup and temperature on the effective thermal conductivity. The effective thermal conductivity of typical irradiated FCM fuel is shown in Fig. 23, the reference value is the thermal conductivity at 600 K, 0% burnup (calculated value 11.35 W/m K). The results indicate that the effect of burnup and temperature on thermal conductivity is at an equal degree. However, Fig. 23(b) shows the effective heat transfer coefficient increases with the increase of TRISO phase volume, which is contrary to the results of unirradiated fuel. This phenomenon should be caused by fission gas release, because of the low thermal conductivity of fission gases, the dominant role of matrix materials is replaced. Also, the decreasing trend of thermal conductivity can be predicted based on the linear increment trend in Fig. 23.

6. Conclusions

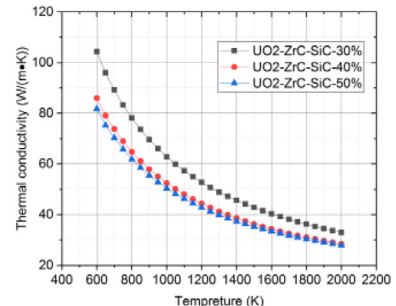
In this paper, a multidimensional multiphysics fuel performance analysis technique for TRISO particle fuel has been established based on the framework of COMSOL Multiphysics. A few fuel behavior models including temperature and burnup dependent properties, fission product swelling, fuel densification, thermal and irradiation creep and fission gas behavior have been integrated into COMSOL user defined modules. Thermomechanical performance, fission gas behavior, failure mechanisms and effective thermal conductivity have been analyzed using the current modified analysis technique. Results of this study can be summarized as follows:



(a). UO_2 -SiC-NITESiC



(b). UO_2 -SiC-Zr



(c). UO_2 -ZrC-NITESiC

Fig. 22. Effective thermal conductivity of unirradiated fuel with temperature.

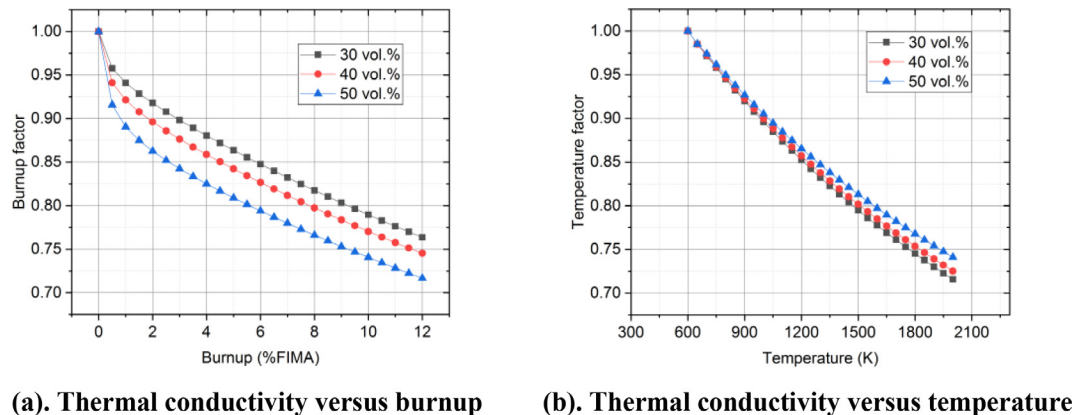


Fig. 23. Effective thermal conductivity of typical irradiated FCM fuel.

1. A precise fission gas behavior considering the grain boundary diffusion model is implemented in this work. Comparisons with experimental measurements and numerical calculation illustrate that this model could successfully predicted fission gas swelling and release at both grain size and fuel particle pellet scales. At the same time, the adopted model will slightly affect the proportion of FGR and gaseous swelling.
2. Thermomechanical analysis of pressure vessel layer in a TRISO particle fuel shows the similar performance of both SiC and ZrC materials. The SiC and ZrC materials were proved to have extremely high structural stability under irradiation. Based on the preliminary thermomechanical analysis, in terms of structural strength and heat transfer efficiency, ZrC appears to be a candidate material of SiC.
3. The kernel migration itself seems to be not enough to break through the porous Buffer or IPyC and destroy the structural integrity of the particle, for no noticeable impact on stress distribution of SiC or ZrC is observed. However, geometric defects such as particle asphericity could be a significant cause to fuel failure.
4. The effective thermal conductivity of different fuel types with different fuel and matrix was analyzed, which is expected to provide a fundamental basis for further precise simulation of the in-pile thermo-mechanical coupling behavior in a fuel pellets level. The results indicate that the effect of burnup and temperature on thermal conductivity is at an equal level. Also, the method and the result could be used in the calculation of fuel pellets scale as an input data.
5. The fuel performance analysis technique in this work is intended as an exploratory assessment of TRISO fuel performance simulation to optimize fuel design. This work may serve as a stepping stone toward a more complete understanding of the TRISO particle fuel. Further works, including transient accident analysis, crack propagation, multiphysics coupling at fuel pellets scale, more mechanism model on Pd corrosion, random dispersion algorithm for TRISO particles are required to be carried out.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.anucene.2020.107599>.

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CRediT authorship contribution statement

Cheng Zhang: Validation, Formal analysis, Writing - original draft. **Yingwei Wu:** Conceptualization, Methodology, Investigation, Writing - original draft. **Shichao Liu:** Writing - original draft. **Ping Chen:** Resources. **Chaoqun Deng:** Formal analysis. **Jing Zhang:** Writing - review & editing. **G.H. Su:** Supervision. **Suizheng Qiu:** Conceptualization, Supervision. **Junmei Wu:** Supervision.

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